

Plant that can move itself to a preferred light source

A team of researchers at Massachusetts Institute of Technology (MIT) Media Lab has built a cyborg that combines a plant with electronics and allows the plant to choose when it would like

to move to a brighter spot. The cyborg is the brainchild of team leader Harpreet Sareen, who has named it Elowan.

Elowan is an attempt to demonstrate what augmentation of nature could mean. Elowan's robotic base is a new symbiotic association with a plant. The agency of movement rests with the plant based on its own bio-electrochemical signals, the language interfaced here with the artificial world. These, in turn, trigger physiological variations such as elongation growth, respiration and moisture absorption.

In this experimental setup, electrodes are inserted into the regions of interest (stems and ground, leaf and ground). Weak signals are then amplified and sent to the robot to trigger movements in respective directions.

Such symbiotic interplay with the artificial world could be extended further with exogenous extensions that provide nutrition, growth frameworks and new defence mechanisms.



Elowan—the cyborg plant—uses its own internal electrical signals to interface with a robotic extension that drives it towards light (Credit: MIT Media Lab)

Fingernail sensors and AI can help monitor health and disease progression

IBM researchers have developed a first-of-a-kind fingernail sensor prototype to help monitor human health. The wearable wireless device continuously measures how a person's fingernail bends and moves, which is a key indicator of grip strength.

Comprising skin, nails and hair, the integumentary system covers most of the human body. Its primary purpose is to protect the internal components from pathogens, toxins, ultraviolet radiation, dehydration and changes in temperature. It also provides a structure for sensory receptors of the somatosensory system of neurons across the body.

One way to measure a disease's progression is to attach skin-based sensors to capture things



Fingernail sensor that can monitor human activities/health (Credit: IBM Lab)